Concept Note
DYNADIST-NA

Dynamical distance optimization for basin-scale
North Atlantic sea-surface temperature jump predictability

Lead institutions: LOPS (Plouzané) and IMT Atlantique / Lab-STICC (Brest)
Scientific lead: Paul Platzer
Local collaborators: Florian Sévellec, Pierre Tandeo
International collaborator: Kinya Toride (Cooperative Institute for Research in Environmental Sciences, University of Colorado Boulder)
Project duration: June 2026 – September 2027

1. Scientific context and motivation

The project addresses a concrete physical question for ocean and climate science: *which oceanic precursors (physical variables and spatial structures) control the predictability of basin-scale North Atlantic sea-surface-temperature (SST) warming transitions at seasonal-to-interannual time scales ?* We focus on a basin-scale **North Atlantic SST anomaly jump** index, defined as the difference between a basin-mean SST anomaly averaged around **+1 year** and around **-10 years** relative to a target date, over a broad North Atlantic domain. This quantity is designed to capture abrupt basin-scale SST transitions while remaining simple enough for systematic analog-based prediction experiments.

The extremely warm North Atlantic conditions of summer 2023, and to a lesser extent 2024, provide a strong motivation for such a framework. The event has been documented extensively, but most studies have emphasized **atmospheric drivers**, exceptional surface forcing, or the combination of internal variability and climate change, while oceanic precursors remain much less systematically explored [1–11]. At the same time, several of these studies suggest that **ocean preconditioning**, including mixed-layer depth and subsurface structure, likely matters for the onset and persistence of such extreme warm states. This makes the North Atlantic an especially relevant testbed for a targeted search for **oceanic sources of predictability**.

The 2023–2024 event does not fit naturally within the most usual definition of a marine heatwave as a relatively **local** and **high-frequency** extreme event [12–14]. Instead, it is better described as a **basin-scale, multi-month to multi-year warm transition**. This justifies the use of a new large-scale index adapted to the question of **slow oceanic preconditioning and predictability**, rather than relying only on standard marine heatwave event definitions tailored to shorter-lived anomalies.

More broadly, recent work increasingly indicates that **ocean dynamics** plays an important role in shaping marine heatwaves and their predictability [15–17]. This is consistent with a long tradition of work showing that the ocean can be a source of predictability from **seasonal** to **interannual** and even **decadal** time scales in the North Atlantic [18–24]. In other words, if a large-scale event such as the 2023 transition does contain predictable components, part of that information may well reside in the slowly evolving ocean state.

This question is tightly connected to the long-standing debate on the respective roles of **atmospheric forcing**, **ocean dynamics**, and **external forcing** in North Atlantic multidecadal SST variability, often referred to as AMV/AMO. The literature contains strongly contrasting interpretations, ranging from dominant roles for aerosols or atmospheric variability to central roles for ocean circulation and heat-content dynamics [25–46]. The present project does not aim to resolve the debate on AMV; instead, it leverages it as a scientific background motivating a practical question: *how much of a basin-scale SST jump can be anticipated from the ocean state alone, and which oceanic processes carry that information?*

2. Observational and simulated datasets

A first step toward this question must rely on **observational datasets**, because they provide the most direct link to the physical system. Several complementary products are available to reconstruct North Atlantic SST and subsurface variability, including EN4-type temperature and salinity fields and long SST analyses such as HadISST, ERSSTv5, HadSST4, and Kaplan SST [47–52]. However, these products also come with well-documented limitations related to sampling, changing observing systems, and historical biases [53–57]. For a question that explicitly concerns **rare, basin-scale, slow transitions**, these limitations matter: the observational record is relatively short, coverage varies strongly across periods, and the number of available extreme transitions is small.

For this reason, the project proposes to use complementary **long climate-model simulations**, especially **CMIP6 coupled runs**, in order to better sample the variability of interest. This strategy follows recent work showing that long archives from coupled models are useful to study multi-year predictability, North Atlantic variability, and data-driven forecast strategies in settings where observations alone remain too short [58–62]. In the present project, model-catalogs allow to increase the sampling of rare trajectories and to test the robustness of identified precursor patterns across data sources.

3. Methodological development

The methodological core of the project is the use of **analog forecasting with optimized distances**. This choice is motivated by both scientific and practical reasons. Analog methods are attractive because they remain **interpretable by construction**: they make predictions by comparing the current state with past states and their subsequent evolutions, rather than by relying on opaque parametric data-driven models. In addition, their behavior is closely tied to the local geometry and density of the data manifold, which makes them particularly suitable for studying **sources of predictability** and their dependence on the available archive [63–65].

At the same time, analog methods are only as good as their **distance function**. The literature has shown repeatedly that the choice or optimization of similarity measures can strongly affect forecast quality [66–69]. In recent work, Platzer et al. (2025) [69] showed that analog distances can be optimized **directly** through a differentiable analog-forecast loss, making it possible to learn metrics tailored to the forecast target and score. This is precisely the methodological direction that motivates the present project.

A second key ingredient comes from the recent work of Toride et al. (2025) [70], who introduced a **state-dependent weighting strategy** using deep learning to identify which regions matter most for SST forecasts in the tropical Pacific ocean. Their results suggest that analog selection should not rely on a single static metric, but on weighting maps that change with the initial state. However, their optimization remains **indirect**: the learned map is not optimized directly through the analog forecast score itself. The main innovation of the proposed DYNADIST-NA project is to combine these two lines of work and test a new **best-of-both-worlds** approach: **state-dependent weights optimized directly for analog forecast performance**. In practice, the project will compare:

1. an **indirect optimization** of state-dependent weighting,
2. a new **direct optimization** of state-dependent analog forecast.

The goal is not only to improve skill, but also to determine whether direct optimization yields weight maps that are more physically meaningful as **precursor diagnostics**.

This makes analog forecasting with optimized, state-dependent distances a particularly judicious tool for the present application. It is sufficiently flexible to learn lead-time-dependent oceanic precursors, sufficiently interpretable to check the physical consistency of the results, and sufficiently lightweight to be deployed quickly on both observation-based and model-based datasets.

4. Project objectives

The project has two tightly coupled objectives.

4.1. Objective 1 – Sources of predictability

Identify the **sources of predictability** of basin-scale North Atlantic SST anomaly jumps at **seasonal to interannual lead times**, using **ocean** predictors. The main outputs will be:

- **state-dependent weight maps** showing which regions and variables matter most for predictability at a given lead time and state,
- a **shortlist of physically interpretable precursor patterns**,
- and an assessment of how these patterns change across lead times.

4.2. Objective 2 – Methodological development

Develop and benchmark a new analog forecasting framework based on **state-dependent direct distance optimization**. The project will compare indirect optimization with a new state-dependent distance **directly optimized for analog forecasting performances**.

The ambition is not only to improve predictability skill, but also to understand **what direct optimization changes physically** in the analog selection process.

5. Methodological plan

The project will rely on **ocean-only data catalogs**, combining:

- **observation-based products** coupling surface SST and subsurface information (e.g. EN4-type products),
- and **long coupled-climate simulations** (CMIP6-like), which provide richer surface + subsurface information and longer training catalogs.

The target variable will be the **continuous SST jump index**, with event-based threshold diagnostics used as a secondary product. Different **lead times** will be treated through **separate optimizations**, since one key scientific hypothesis is that different predictors and regions control different time scales of predictability.

The methodological comparison is the core of the project.

5.1. Method A – Indirect state-dependent weighting

Reproduce and adapt the first approach [70], in which a model learns **state-dependent weighting maps** that affect analog selection, but are not optimized directly through the analog forecast score.

5.2. Method B – Direct state-dependent optimization

Develop a new approach in which the state-dependent weighting maps are optimized **directly** by backpropagating through a differentiable analog-forecast loss. The analog forecast remains entirely data-driven, and no differentiation through a climate model is required.

5.3. Evaluation

Both methods will be compared on:

- forecast skill at different lead times,

- robustness across observational and model-based catalogs,
- and interpretability of the learned weight maps.

The physical interpretation will focus on whether the learned maps identify **coherent precursor regions and structures** in the North Atlantic ocean state, and whether those structures differ between **seasonal** and **interannual** predictability horizons.

6. Work plan and deliverables

6.1. Phase 1 (mid-2026): dataset preparation and protocol

- Assemble the first observational and modeled datasets.
- Compute the associated SST jump index.
- Set the verification protocol (skill score) and a baseline unoptimized analog method.

6.2. Phase 2 (beginning-2027): benchmarking

- Implement the baseline state-dependent weighting approach of Toride et al. (2025) [70].
- Perform benchmark experiments across several lead times.
- Prepare internship tasks and technical pipeline.

6.3. Phase 3a (March-August 2027, M2 internship): implementation and experiments

The project will support **one M2 internship** in 2027. The intern will play a central role in:

- implementing the new direct optimization method,
- running ablation studies and hyperparameter sensitivity tests,
- comparing direct versus indirect state-dependent optimization,
- writing a paper for AMS journal *Artificial Intelligence for the Earth Systems*.

6.4. Phase 3b (April 2027): international exchange

The project includes a **2 week research visit of Kinya Toride (Cooperative Institute for Research in Environmental Sciences, University of Colorado Boulder) to Brest**, during which:

- guidance to the intern will be provided along with scientific discussion
- and a **one-day seminar** will be organized with PORC-EPIC participants and other external colleagues working on predictability, analog forecasting, and extremes.
- future collaborations between ISblue-laboratory members and CIRE/NOAA members will be discussed

6.5. Phase 4 (September 2027): cementing future collaborations

Further collaborations will be cemented by preparing submissions to larger funding schemes (*e.g.* INSU LEFE/SUN; ANR).

6.6. Main deliverables

- a reproducible, **open-source python code**,
- a comparison between indirect and direct state-dependent analog optimization,
- lead-time-dependent weight maps and **precursor diagnostics** for North Atlantic SST jumps,
- a **scientific paper** published in American Meteorological Society (AMS) journal *Artificial Intelligence for the Earth Systems*
- a short report preparing larger follow-up proposals.

7. Team and collaborations

The project is built around a strong **Brest-based interdisciplinary core**:

- **Paul Platzer** (IMT Atlantique / Lab-STICC; scientific lead; analog forecasting, distance learning, predictability),
- **Florian Sévellec** (CNRS / LOPS; North Atlantic variability and predictability),
- **Pierre Tandeo** (IMT Atlantique / Lab-STICC; data assimilation, analog methods, AI for geosciences).

Its main **international asset** is the collaboration with:

- **Kinya Toride** (Cooperative Institute for Research in Environmental Sciences, University of Colorado Boulder), whose recent work on state-dependent weighting maps is one of the direct starting points of the project.

This collaboration is a strong match for **ISblue's structuring and internationalization goals**, as it combines (i) a concrete scientific application of direct relevance to ocean and climate research, (ii) a methodological collaboration with an international partner, and (iii) a local interdisciplinary framework bringing together LOPS and IMT Atlantique / Lab-STICC. Through the M2 internship, the reproducible code and notebooks, and the one-day seminar, the project will generate reusable tools, strengthen exchanges within the ISblue community, and help position ISblue as a visible hub for interpretable data-driven approaches to ocean predictability and extremes.

8. Requested support

The requested ISblue support will cover:

- **one M2 internship** (core implementation and experiments),
- **international collaboration support** for a 2 week research visit of Kinya Toride in Brest,
- **low-cost scientific animation** for a one-day seminar,

The project is intentionally **compact, low-cost, and high-leverage**: it supports a targeted scientific question, a strong methodological innovation, a training component, and a visible international collaboration.

Publication fees associated with the project will be covered by PORC-ÉPIC ANR led by Florian Sévellec, which has many connections with the proposed DYNADIST-NA project.

More details on the project budget is given in a dedicated document.

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